University of Buea
Department of Computer Science
DCS602: Information Visualization

Homework 1

Deadline for Submission: 15th May, 2026

Objectives

1. An overview of various basic visualization techniques
2. Familiarization with the Matplotlib visualization library

Dataset: The old cars dataset.
The dataset lists several characteristics of various cars built between 1970 and 1982. The dataset is typical tabular dataset with items corresponding to the individual car models and the attributes correspond to various characteristics. The attributes are:

a) Car: Car name and model
b) MPG (Miles per Gallon): The fuel efficiency - how many miles a car can travel on one gallon of fuel
c) Displacement: The total volume of an engine's cylinders, usually measured in cubic inches and indicating the engine size and capacity
d) Horsepower: The power produced by the engine to move the car
e) Weight: The weight of the vehicle, measured in pounds
f) Model: The year of release
g) Origin: The manufacturing region of the car

Task 1: Grouped bar chart
Visualize with a grouped bar chart the gas mileage distribution of all the models by geographic origin, i.e, a chart in which each gas mileage value is associated with 3 bars (one for each origin: US, Europe, Japan), showing how the distributions of the gas mileage values compare across regions. Note: Discretize the range of possible gas mileage values in 2 mpg increments.
Deliverable: pr1-groupedBar.py

Task 2: Line Chart
Use a line chart to visualize for each geographic origin the evolution of the gas mileage over the years from 1970 to 1982. Each data point will correspond to the annual average of the gas mileage for a given origin and each curve will comprise 13 points. Assign a different colour to each curve.
Deliverable: pr1-line.py

Task 3: Scatter plot
Use a scatter plot to visualize the relationship between horsepower and gas mileage. The horizontal axis should correspond to the horsepower, the vertical axis to the gas mileage and each data point to a particular car. Use colour coding to indicate the year.
Deliverable: pr1-scatterplot.py

Task 4: Task 3: Scatter plot matrix
Visualize gas mileage, weight, horsepower, and engine size in a scatter plot matrix. Colour code the individual data points by country of origin.
Note: Matplotlib does not have a built-in "scatter plot matrix" function in its core library.
Investigate how you can use pandas.plotting.scatter_matrix to achieve this.
Deliverable: pr1-scattermatrix.py